

ROTUMA ISLAND, Fiji

WMO: 916500

Lat: 12.500S

Lon: 177.050E

Elev: 85

StdP: 14.65

Time Zone: 12.00 (PCF)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	74.9	75.7	67.6	101.6	79.0	69.0	106.8	79.5	18.7	80.7	17.4	78.0	4.3	90	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.0	89.0	80.5	88.3	80.2	87.8	80.0	82.0	86.8	81.5	86.4	80.9	86.1	5.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.7	160.0	85.7	80.0	156.4	85.2	79.2	152.2	84.6	46.0	86.9	45.3	86.7	44.8	86.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.0	12.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	82.6	82.7	82.5	82.7	81.9	81.5	81.1	80.5	81.0	81.2	81.8	82.4	
	DBStd	1.77	1.82	1.56	1.61	1.49	1.57	1.57	1.47	1.57	1.70	1.59	1.62	1.72	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	11610	1010	916	1008	980	988	944	963	945	930	968	955	1003	
	CDD65	6135	545	496	543	530	523	494	498	480	480	503	505	538	
	CDH74	65824	6215	5675	6146	5961	5587	5100	5006	4599	4830	5208	5452	6045	
	CDH80	17821	2040	1844	1955	1871	1496	1170	1026	824	993	1227	1481	1894	
Wind	WSAvg	4.6	4.6	4.0	4.2	3.7	3.5	4.7	5.2	5.1	5.8	4.6	4.7	4.4	
Precipitation	PrecAvg	136.8	13.7	12.7	13.7	10.9	11.1	10.3	9.2	8.4	8.5	13.4	12.2	12.2	
	PrecMax	172.4	37.6	29.6	39.6	22.5	20.6	21.8	18.5	23.7	17.5	25.8	24.0	26.1	
	PrecMin	96.5	1.1	4.4	3.8	2.4	1.3	0.7	1.5	0.9	1.9	4.9	4.2	1.1	
	PrecStd	19.0	6.1	4.8	7.2	4.7	4.6	5.0	4.7	4.9	3.8	5.6	4.3	5.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	89.6	89.7	89.3	88.5	87.4	87.5	87.1	87.1	87.6	88.4	89.3	
		MCWB	80.9	81.0	80.9	80.7	80.5	79.8	79.5	79.7	78.7	79.4	79.6	80.5	
	2%	DB	88.7	88.4	88.4	88.2	87.3	86.5	86.3	86.0	86.2	86.6	87.3	88.2	
		MCWB	80.5	80.3	80.3	80.2	80.0	79.3	78.8	78.8	78.7	79.0	79.4	80.0	
	5%	DB	87.9	87.6	87.6	87.3	86.5	85.8	85.3	85.0	85.5	85.9	86.6	87.4	
		MCWB	80.1	80.0	80.0	79.8	79.6	78.9	78.4	78.3	78.4	78.7	79.1	79.7	
	10%	DB	87.0	86.8	86.7	86.5	85.7	85.0	84.4	83.9	84.7	85.1	85.9	86.4	
		MCWB	79.8	79.6	79.7	79.5	79.1	78.5	77.9	77.8	78.1	78.2	78.7	79.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.4	82.4	82.0	82.4	81.0	81.0	80.8	80.6	81.0	81.9	82.3
			MCDB	87.4	87.7	87.4	87.1	86.2	85.7	85.6	85.8	85.0	85.8	86.0	86.9
2%		WB	81.7	81.2	81.4	81.0	81.2	80.4	79.9	79.9	79.7	80.0	80.9	81.2	
		MCDB	86.8	86.8	86.6	86.4	85.5	85.3	84.4	84.5	84.3	85.1	85.5	86.3	
5%		WB	81.0	80.6	80.8	80.6	80.6	79.6	79.1	79.0	79.1	79.2	80.1	80.5	
		MCDB	86.3	86.5	86.1	86.1	85.2	84.3	83.6	83.4	83.7	84.2	84.8	85.9	
10%		WB	80.4	80.0	80.1	80.0	79.8	79.0	78.5	78.3	78.6	78.7	79.4	79.9	
		MCDB	85.7	85.5	85.3	85.2	84.3	83.5	82.9	82.5	83.1	83.6	84.1	85.0	
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.0	8.0	7.7	7.4	6.7	6.7	6.6	6.4	7.4	7.5	8.0	
		MCDBR	9.3	9.2	9.2	8.7	8.5	7.7	7.9	8.2	7.5	8.4	8.6	9.3	
		MCWBR	4.7	4.5	4.4	4.4	4.2	4.2	4.0	4.2	3.6	4.3	4.5	4.8	
	5% WB	MCDBR	8.5	8.8	8.6	8.2	8.1	7.3	7.4	7.6	7.1	8.0	8.1	8.8	
		MCWBR	4.8	4.6	4.5	4.5	4.5	4.3	4.1	4.3	3.9	4.5	5.0	5.0	
Clear-Sky Solar Irradiance	taub	0.409	0.405	0.395	0.394	0.401	0.420	0.421	0.418	0.422	0.426	0.417	0.416		
	taud	2.559	2.576	2.608	2.613	2.583	2.497	2.494	2.504	2.484	2.488	2.525	2.534		
	Ebn at Noon	297	296	293	283	271	259	262	272	280	286	292	294		
	Edh at Noon	35	34	32	30	30	31	32	33	36	36	36	35		
All-Sky Solar Radiation	RadAvg	1805	1745	1777	1618	1442	1344	1423	1608	1785	1831	1834	1826		
	RadStd	228	182	176	157	118	84	79	84	93	107	139	109		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page